Does the regime classifier add value?

Short answer

The regime flag adds a small amount of ranking power in both model families, but the gains are within what a placebo flag with the same persistence produces. The evidence that the regime classifier adds out-of-sample value is weak. The flag also worsens calibration in 2023. The choice of model matters more than the flag, and which model wins depends on the period.

1. Value of the regime flag (M2 vs M1)

- 2022 validation AP: logit 0.30 -> 0.36 (+0.055); RF 0.57 -> 0.64 (+0.070). Both gains have bootstrap intervals spanning zero by a wide margin and placebo p = 0.22 (logit) and 0.17 (RF). Only 8 breaches, all in July 2022.
- 2022 validation Brier: the flag improves calibration for both models; for RF this beats almost every placebo (p = 0.011), for logit p = 0.11. This is the strongest pro-regime result, It comes from July and August, where M2 lowers false alarms; in September (mostly stressed, no breaches) M2 is worse for both models.
- 2023 frozen test AP: logit +0.028 (bootstrap CI [-0.005, +0.055], placebo p = 0.11); RF +0.014 (placebo p = 0.16). Logit AUC gain +0.019 is the only interval excluding zero (placebo p = 0.06).
- 2023 Brier gets worse with the flag (logit +0.0211, CI excludes zero; RF +0.0046). Reason: the flag is on for only 3 days in 2023. M2 learned from 2022 that breaches cluster in stress, so it cuts normal-day probabilities (logit mean 8.5% -> 3.0%, RF 10.6% -> 7.0%), but 40 of the 49 2023 breaches happened on normal-regime days.
- Lag sensitivity: the 2022 AP gain moves a lot with the flag's timing (largest at lag 1 for both models) while 2023 barely changes. A real signal should not hinge on a one-day shift; this points to noise from the few July 2022 breaches.

2. Logit vs random forest

- 2022 validation: RF ranks breaches much better (M1 AP 0.57 vs 0.30) and is better calibrated (Brier 0.038 vs 0.052).
- 2023 frozen test: logit generalises better (M2 AP 0.56 vs 0.51; AUC 0.84 vs 0.75). The forest fits 2022 patterns closely and extrapolates poorly to the 2023 regime of on-chain activity; the logit's smooth slopes carry over better.
- On the SVB stressed days the flag adds about +3 percentage points for USDC/DAI in both M2s. For logit this lifts M2 above M1; for RF, M2's stress-off baseline is lower, so RF M2 ends up roughly equal to RF M1. Neither model gives DAI or USDC a high probability during SVB (RF peaks near 25% for USDC, logit near 6%); both mainly flag USDT.

3. Does the effect of on-chain activity depend on the regime?

Does the regime classifier add value? (cont.)

- Yes, in-sample, in both families. Logit M3 Wald p = 2.3e-10 (but with a sparse-data warning). RF regime-dependence of partial dependence beats placebos (p = 0.033), driven by burned, minted and unique receivers. This heterogeneity does not turn into better out-of-sample forecasts.

Bottom line for the paper

Report the regime flag as a descriptive moderator (on-chain effects differ under stress), not as a forecasting improvement. Out of sample, the flag's gains are small, timing-sensitive and indistinguishable from placebo flags, and the classifier marked almost none of the 2023 depeg period as stressed. RF wins in 2022 and logit wins in 2023, so neither model choice changes this conclusion.

Setup

Question

Two nested specifications are compared within each model family. M1 uses seven log1p on-chain features (transfers, volume, unique senders/receivers, minted, burned, large-transfer volume) for coin c on day t. M2 adds the binary market-stress flag from regime_classifier for day t. The target is depeg_dynamic on day t+1. The regime classifier 'adds value' if M2 forecasts breaches better than M1 out of sample.

Models

- Logit: sklearn LogisticRegression, C = 1 (L2), liblinear, features standardised on training data only. Identical to logit/logit_oos.py (re-run; results match the saved run).
- Random forest: 500 trees, max_features = sqrt, min_samples_leaf = 5, bootstrap, no class weights, random_state = 0, no scaling. Fixed in advance, no tuning, mirroring the logit's fixed C. Code: random_forest/rf_oos.py.

Evaluation design (identical for both models)

- Panel: 3 coins (DAI, USDC, USDT); 2022 predictor days Apr to Nov, 2023 predictor days Feb to Apr.
- 2022 sequential validation: monthly folds Jul to Nov 2022, each trained only on outcomes before the fold starts (expanding window, at least 90 training dates). 372 forecasts, 8 breaches, all in July.
- 2023 frozen test: models trained on all of 2022 and frozen; no 2023 labels are used. 267 forecasts, 49 breaches, but only 3 stressed days (all in the SVB week).
- Metrics: average precision (AP, main metric for rare events), ROC AUC, Brier score (lower is better). M2 minus M1 is the value added by the flag.

How 'value added' is tested

- Block bootstrap: dates are resampled in 7-day blocks (coins on a date stay together) to give a 95% interval for M2 minus M1.
- Placebo regimes: the real stress series is circularly shifted along its own calendar by 30 to 646 days (89 shifts, step 7). A placebo keeps the flag's share of stressed days and spell lengths but breaks its timing. Each placebo goes through the full pipeline. If the real flag carries information, its M2 gain should beat most placebo gains. p = share of placebos at least as good.
- Interaction test: does the effect of on-chain activity differ by regime? Logit: M3 with seven feature-by-stress terms and a date-clustered Wald test (logit/interaction_test.py). RF: the forest learns interactions implicitly, so we measure how much each feature's partial dependence curve changes shape between stress = 0 and 1, versus placebo flags (random_forest/rf_interaction_test.py).

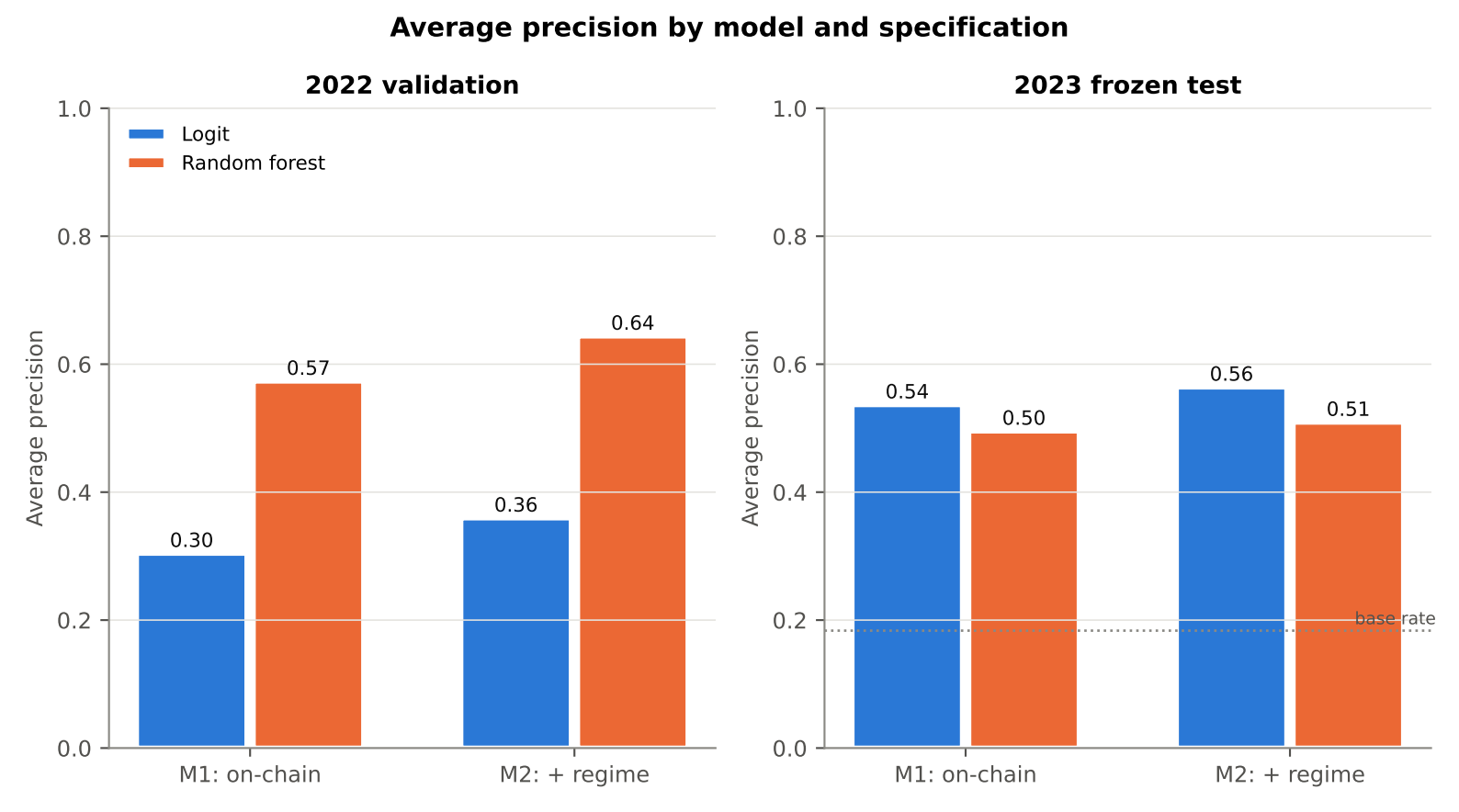
Out-of-sample metrics: M1 (on-chain) vs M2 (+ regime flag)

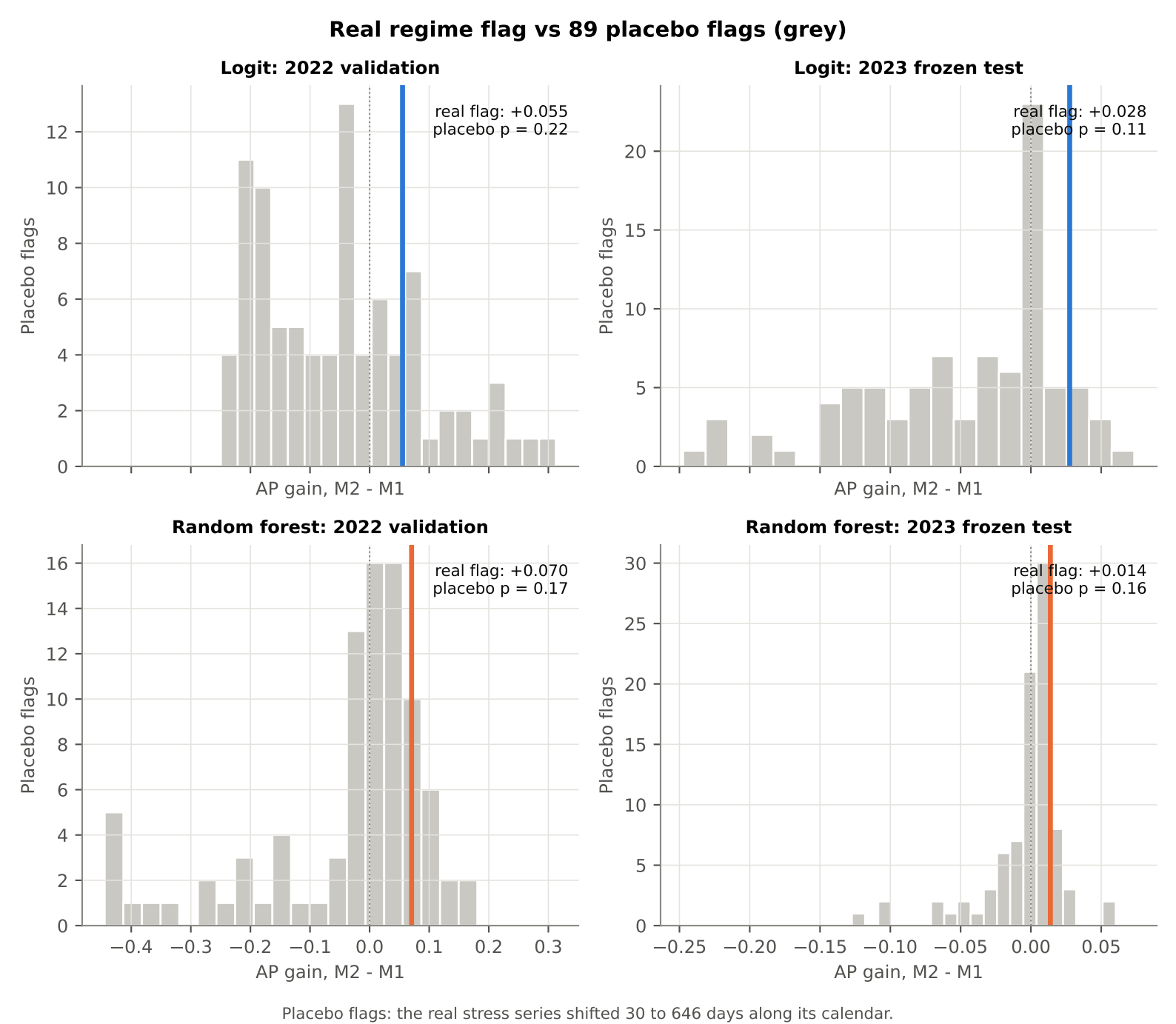
Phase	Model	Spec	n	Breaches	AP	ROC AUC	Brier
2022 validation	Logit	M1	372	8	0.304	0.958	0.0516
2022 validation	Logit	M2	372	8	0.360	0.945	0.0478
2022 validation	Random forest	M1	372	8	0.573	0.979	0.0380
2022 validation	Random forest	M2	372	8	0.644	0.971	0.0331
2023 frozen test	Logit	M1	267	49	0.537	0.826	0.1310
2023 frozen test	Logit	M2	267	49	0.564	0.845	0.1521
2023 frozen test	Random forest	M1	267	49	0.495	0.739	0.1317
2023 frozen test	Random forest	M2	267	49	0.509	0.753	0.1363

Value added by the regime flag (M2 minus M1). Brier: negative = M2 better.

Phase	Model	Metric	M2 - M1	95% bootstrap CI	M2 better	Placebo p
2022 validation	Logit	AP	+0.055	[-0.385, +0.319]	58%	0.22
2022 validation	Logit	AUC	-0.013	[-0.058, +0.056]	42%	0.54
2022 validation	Logit	Brier	-0.0038	[-0.0260, +0.0201]	63%	0.11
2022 validation	Random forest	AP	+0.070	[-0.176, +0.538]	63%	0.17
2022 validation	Random forest	AUC	-0.008	[-0.030, +0.034]	53%	0.74
2022 validation	Random forest	Brier	-0.0049	[-0.0210, +0.0111]	73%	0.01
2023 frozen test	Logit	AP	+0.028	[-0.005, +0.055]	94%	0.11
2023 frozen test	Logit	AUC	+0.019	[+0.002, +0.040]	100%	0.06
2023 frozen test	Logit	Brier	+0.0211	[+0.0009, +0.0466]	2%	0.94
2023 frozen test	Random forest	AP	+0.014	[-0.003, +0.032]	93%	0.16
2023 frozen test	Random forest	AUC	+0.013	[-0.002, +0.026]	96%	0.63
2023 frozen test	Random forest	Brier	+0.0046	[-0.0074, +0.0202]	24%	0.89

M2 better: share of bootstrap draws in which M2 beats M1. Placebo p: share of 89 placebo flags whose M2 gain is at least as good as the real flag's. Small p means the real regime timing matters; p near 0.5 or above means a random flag with the same persistence does as well. The 2022 validation has only 8 breaches (all July), so its intervals are wide.


Average precision by model and specification
2022 validation
2023 frozen test
Logit
Random forest
Average precision
1.0
0.8
0.6
0.4
0.2
0.0
0.30
0.57
0.36
0.64
0.54
0.50
0.56
0.51
base rate
M1: on-chain
M2: + regime


Real regime flag vs 89 placebo flags (grey)
Logit: 2022 validation
real flag: +0.055
placebo p = 0.22
Logit: 2023 frozen test
real flag: +0.028
placebo p = 0.11
Random forest: 2022 validation
real flag: +0.070
placebo p = 0.17
Random forest: 2023 frozen test
real flag: +0.014
placebo p = 0.16
Placebo flags
AP gain, M2 - M1
Placebo flags: the real stress series shifted 30 to 646 days along its calendar.

Where the gains come from

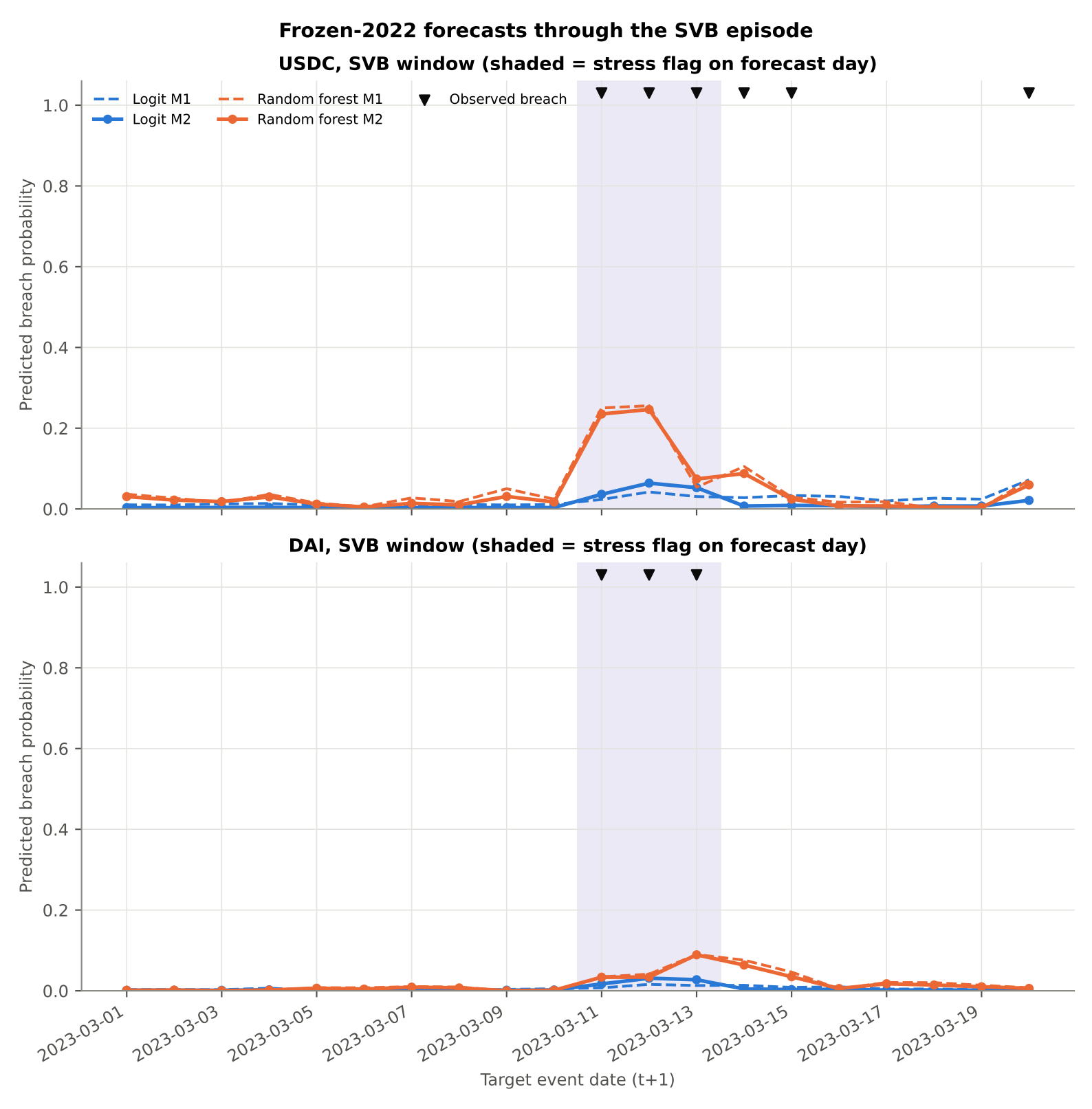
2022 validation by fold (AP undefined without breaches)

Model	Fold	n	Breaches	Stressed	AP M1	AP M2	Brier M1	Brier M2
Logit	2022-07	93	8	27	0.333	0.751	0.0884	0.0483
Logit	2022-08	93	0	9			0.0654	0.0246
Logit	2022-09	90	0	72			0.0345	0.0963
Logit	2022-10	93	0	48			0.0190	0.0252
Logit	2022-11	3	0	0			0.0077	0.0010
Random forest	2022-07	93	8	27	0.621	0.722	0.0722	0.0439
Random forest	2022-08	93	0	9			0.0474	0.0198
Random forest	2022-09	90	0	72			0.0163	0.0561
Random forest	2022-10	93	0	48			0.0145	0.0137
Random forest	2022-11	3	0	0			0.0663	0.0185

Split by the regime on the forecast day

Phase	Model	Regime	n	Breaches	AP M1	AP M2	Brier M1	Brier M2
2022 validation	Logit	normal	216	2	0.212	0.217	0.0600	0.0174
2022 validation	Logit	stressed	156	6	0.815	0.438	0.0399	0.0900
2022 validation	Random forest	normal	216	2	0.542	0.537	0.0488	0.0194
2022 validation	Random forest	stressed	156	6	0.733	0.788	0.0231	0.0520
2023 frozen test	Logit	normal	258	40	0.494	0.498	0.1105	0.1342
2023 frozen test	Logit	stressed	9	9			0.7199	0.6669
2023 frozen test	Random forest	normal	258	40	0.438	0.441	0.1176	0.1222
2023 frozen test	Random forest	stressed	9	9			0.5365	0.5410

Within a regime the M2 flag is constant, so M2 can only re-rank observations through the intercept shift (logit) or through flag-dependent splits (RF). Gains therefore show up mostly in Brier (calibration) and in the pooled ranking across regimes.

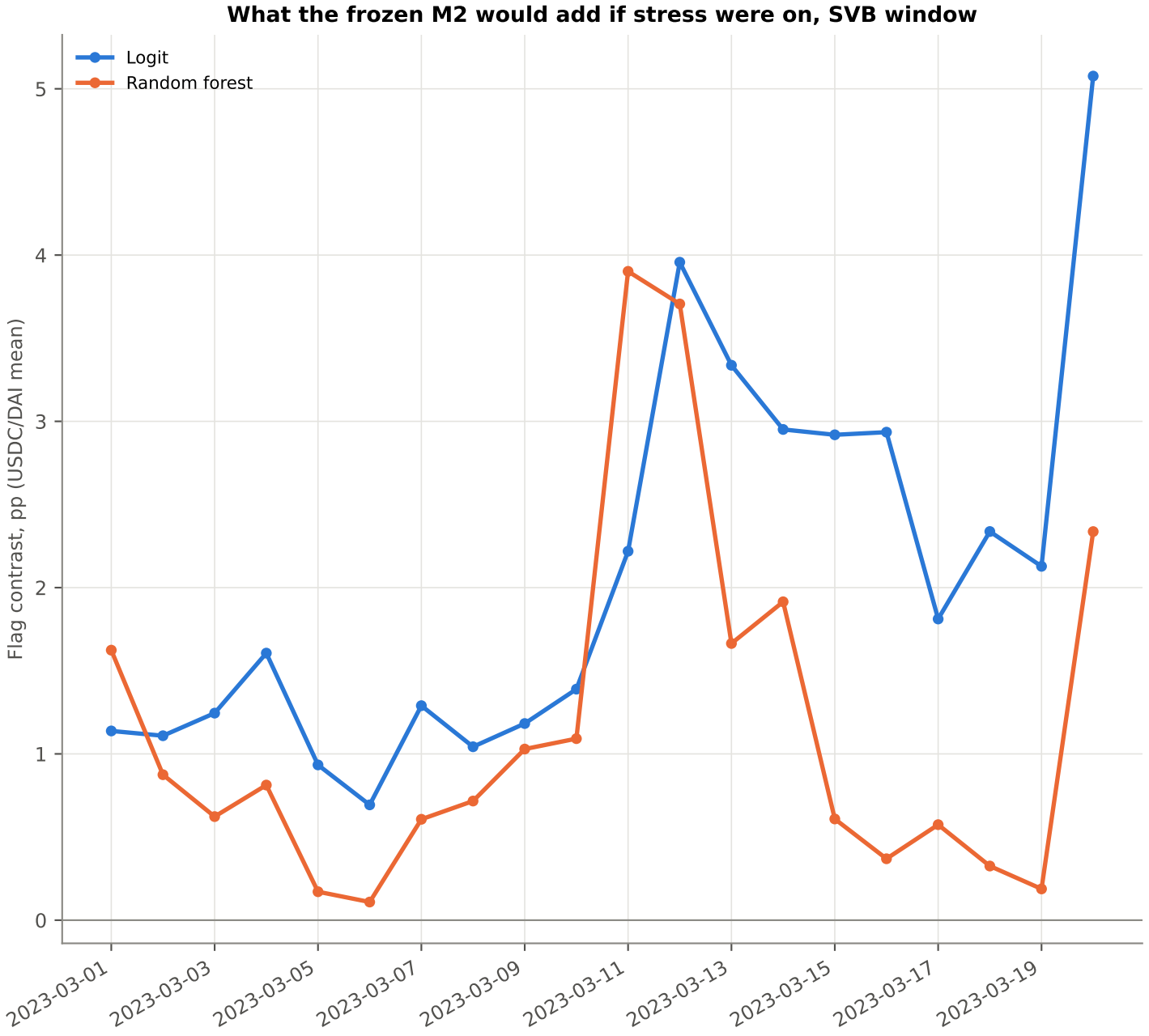

Frozen-2022 forecasts through the SVB episode
USDC, SVB window (shaded = stress flag on forecast day)
Logit M1
Logit M2
Random forest M1
Random forest M2
Observed breach
Predicted breach probability
1.0
0.8
0.6
0.4
0.2
0.0
DAI, SVB window (shaded = stress flag on forecast day)
Predicted breach probability
1.0
0.8
0.6
0.4
0.2
0.0
2023-03-01
2023-03-03
2023-03-05
2023-03-07
2023-03-09
2023-03-11
2023-03-13
2023-03-15
2023-03-17
2023-03-19
Target event date (t+1)

How much of M2's forecast comes from the flag?

Model	Phase	Mean contrast	Median contrast	Contrib. (stressed)	Contrib. (SVB, USDC/DAI)
Logit	2022 validation	+16.0	+5.9	+16.2	
Logit	2023 frozen test	+10.3	+1.8	+15.5	+3.2
Random forest	2022 validation	+9.2	+0.2	+9.2	
Random forest	2023 frozen test	+5.6	+1.3	+11.9	+3.1

All values in percentage points.

Contrast = P(breach | stress = 1) - P(breach | stress = 0) from the same fitted M2 with on-chain inputs held fixed (pp = percentage points). Contribution = actual M2 - M2 with stress forced to 0; it is zero on normal days by construction. Model-based decomposition, not a causal effect.



Regime lag sensitivity (flag from t, t-1 or t-3)

Model	Phase	Lag (days)	AP M1	AP M2	AP gain	Brier M1	Brier M2
Logit	2022 validation	0	0.304	0.360	+0.055	0.0516	0.0478
Logit	2022 validation	1	0.304	0.459	+0.155	0.0516	0.0466
Logit	2022 validation	3	0.304	0.440	+0.135	0.0516	0.0504
Logit	2023 frozen test	0	0.537	0.564	+0.028	0.1310	0.1521
Logit	2023 frozen test	1	0.537	0.561	+0.025	0.1310	0.1535
Logit	2023 frozen test	3	0.537	0.560	+0.024	0.1310	0.1512
Random forest	2022 validation	0	0.573	0.644	+0.070	0.0380	0.0331
Random forest	2022 validation	1	0.573	0.773	+0.199	0.0380	0.0346
Random forest	2022 validation	3	0.573	0.614	+0.040	0.0380	0.0386
Random forest	2023 frozen test	0	0.495	0.509	+0.014	0.1317	0.1363
Random forest	2023 frozen test	1	0.495	0.501	+0.005	0.1317	0.1392
Random forest	2023 frozen test	3	0.495	0.503	+0.007	0.1317	0.1367

On-chain inputs stay at lag 0, so M1 is identical across rows (checked in code). Only the regime flag's timing changes. Lag choice was not tuned on 2023. If the gain depends heavily on whether the flag is from t or t-1, the evidence for the regime signal is fragile.

Do on-chain effects differ by regime? (interaction tests, 2022 sample)

Logit M3: joint Wald chi-square(7) = 59.1, p = 2.3e-10, date-clustered, 645 obs / 215 dates. Warning from the script: one regime has fewer breaches than parameters, so the asymptotic test may be unreliable.

Logit M3: regime-specific log-odds slopes (standardised features)

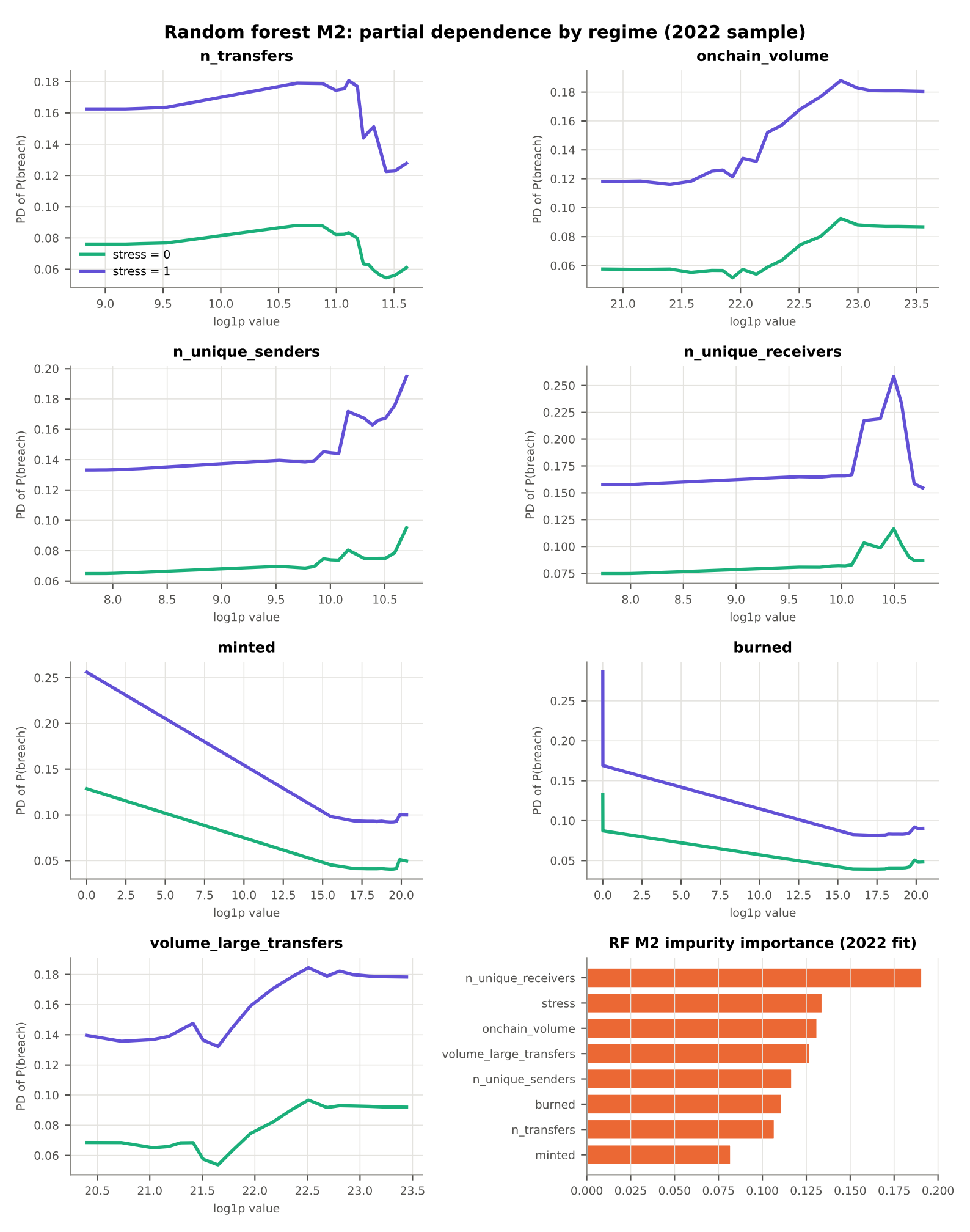
Feature	Normal slope (SE)	Stressed slope (SE)	Interaction
n_transfers	-24.02 (6.60)	-5.56 (4.99)	+18.46
onchain_volume	0.97 (5.43)	8.20 (1.87)	+7.23
n_unique_senders	16.79 (4.43)	3.73 (3.51)	-13.06
n_unique_receivers	10.73 (3.92)	-0.44 (1.85)	-11.17
minted	8.32 (2.91)	-2.33 (1.29)	-10.65
burned	-5.83 (2.21)	-0.41 (1.09)	+5.43
volume_large_transfers	-0.77 (4.64)	-6.27 (1.83)	-5.51

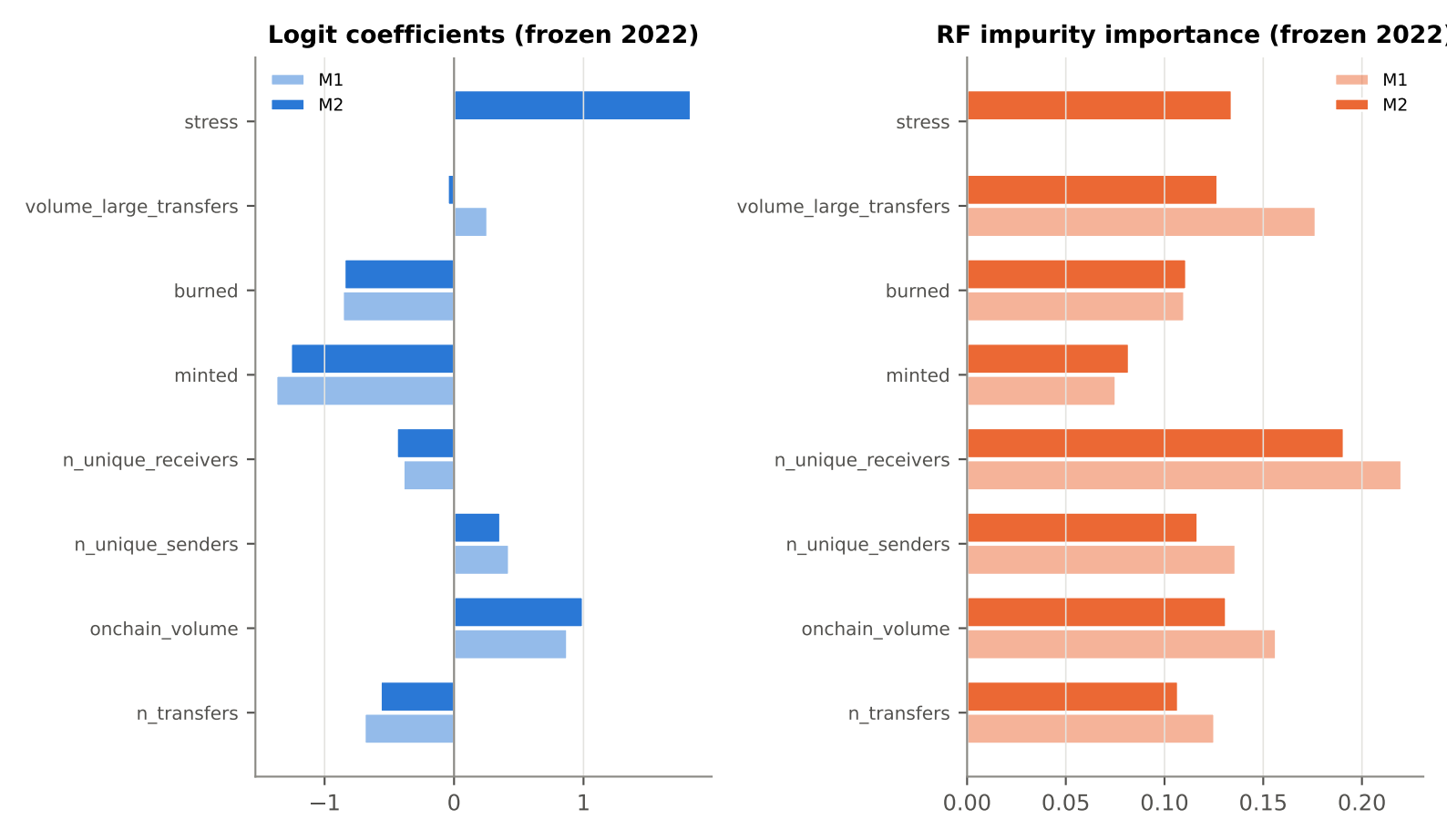
Random forest: joint interaction statistic = 0.110 vs placebo mean 0.048; placebo p = 0.033 (89 placebos). OOB AP: M1 0.645, M2 0.707, gain placebo p = 0.20.

RF M2: regime-dependence of each feature's partial dependence

Statistic	Real	Placebo mean	Placebo 95th pct	Placebo p
joint interaction	0.1097	0.0481	0.1029	0.033
oob ap gain	0.0626	0.0344	0.1097	0.200
n transfers	0.0090	0.0067	0.0157	0.222
onchain volume	0.0142	0.0059	0.0141	0.067
n unique senders	0.0118	0.0061	0.0139	0.089
n unique receivers	0.0196	0.0080	0.0169	0.044
minted	0.0196	0.0066	0.0180	0.011
burned	0.0286	0.0085	0.0246	0.011
volume large transfers	0.0068	0.0062	0.0139	0.378

I_j = standard deviation over a quantile grid of [PD_j(g | stress = 1) - PD_j(g | stress = 0)], on the probability scale. A flat difference means the flag only shifts the level; a varying difference means the feature's effect depends on the regime. Both tests are in-sample descriptions of 2022, not out-of-sample evidence.


Random forest M2: partial dependence by regime (2022 sample)
n_transfers
onchain_volume
n_unique_senders
n_unique_receivers
minted
burned
volume_large_transfers
RF M2 impurity importance (2022 fit)
PD of P(breach)
log1p value
stress = 0
stress = 1
n_unique_receivers
stress
onchain_volume
volume_large_transfers
n_unique_senders
burned
n_transfers
minted


Logit coefficients (frozen 2022)
M1
M2
stress
volume_large_transfers
burned
minted
n_unique_receivers
n_unique_senders
onchain_volume
n_transfers
−1
0
1
RF impurity importance (frozen 2022)
M1
M2
stress
volume_large_transfers
burned
minted
n_unique_receivers
n_unique_senders
onchain_volume
n_transfers
0.00
0.05
0.10
0.15
0.20

Caveats and reproduction

Caveats

- Very few events. The 2022 validation has 8 breaches, all in July 2022; Aug to Nov folds have none, so they contribute only to Brier. Any 2022 out-of-sample conclusion rests on one month.
- The 2023 test has 49 breaches but the regime flag is on for only 3 days (SVB week). M2 and M1 differ only on those days, so the 2023 test is close to a test of 'what happens on 3 days'.
- Placebo flags are overlapping shifts of one series, so they are not independent draws; treat placebo p-values as descriptive. With 89 placebos the smallest attainable p is about 0.011.
- Both models use fixed, untuned specifications. A tuned forest could look different, but tuning on this few events would mostly fit noise.
- Publication timing of the on-chain aggregates and regime inputs is assumed (lag 0), not verified; the lag-sensitivity page shows how much this matters for the flag.

Code and outputs

- logit/: logit_oos.py, regime_lag_sensitivity.py, interaction_test.py (unchanged).
- random_forest/: rf_oos.py, rf_regime_lag_sensitivity.py, rf_interaction_test.py, with outputs in random_forest/outputs/.
- model_comparison/: compare_models.py (bootstrap + placebo for both models) and build_report.py (this PDF).

Run order from the repository root: python random_forest/rf_oos.py --out random_forest/outputs/rf_decomposition; python random_forest/rf_regime_lag_sensitivity.py; python random_forest/rf_interaction_test.py; python model_comparison/compare_models.py; python model_comparison/build_report.py.